Design and Implementation Security Operations Center (SOC) Using Open Source Technologies SIEM for Industries

A project paper is submitted to the **Faculty of Engineering & Technology** of **Islamic University** in partial fulfillment of the requirements for the Degree of B.Sc. Engineering in **Information and Communication Technology**.

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Session: B.Sc. (2019-120)

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

ISLAMIC UNIVERSITY, KUSHTIA, BANGLADESH

CERTIFICATE

This is to certify that the work presented in the project "**Design and Implementation Security Operations Center (SOC) Using Open Source Technologies SIEM for Industries**" is the result of research conducted by **Md Mesbah Uddin** under my supervision in the academic year 2019–2020, belongs to Shahid Ziaur Rahman Hall, examination roll number-**1918010**, registration number-1318, and was submitted in partial fulfillment of the requirements for the degree of B.Sc. Engineering. Additionally, he completed this unique project work as part of his B.Sc. engineering studies at the Department of Information and Communication Technology, Islamic University, Kushtia, Bangladesh.

To the best of my knowledge, this study has not been submitted anywhere else, nor has it been copied from any other research prior to being presented to this department.

Signature of the Supervisor

……

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Abstract

In today's rapidly advancing technological landscape, safeguarding sensitive data has become a major challenge for organizations of all scales, including industries, enterprises, and startups. The rise of sophisticated cyber threats and the availability of advanced hacking tools have made it increasingly difficult to ensure robust information security. Over the past few years, large industries in particular have experienced a growing number of security breaches, raising serious concerns about data protection. To counter these risks, businesses are seeking innovative approaches to strengthen their defenses against cyberattacks. One effective solution is the establishment of a Security Operations Center (SOC), which serves as a central facility for monitoring, detecting, preventing, and responding to malicious activities within a network. SOC play a vital role in shaping cybersecurity strategies by providing in-depth analysis and actionable insights into hostile activities. However, in many organizations, the SOC is still treated as an optional component rather than a core part of IT infrastructure. The objective of this project is to evaluate open-source SOC tools for industrial environments and to design an operational framework that leverages open-source SIEM technologies for effective security monitoring

Proposed method:

When an attacker attacks, We can capture it from the on a switch or NIC card and monitor it in real time before it enters our system.We will use Suricata tool for monitoring and capture the network packets. Suricata operates in promiscuous mode, enabling its network interface card to capture and analyze all traffic passing through the switch

Acknowledgement

First and first, I would want to express my sincere appreciation to Allah, the Almighty, the Great Creator, for providing me the chance to work on and successfully complete the project dubbed "Design and Implementation Security Operations Center (SOC) Using Open Source Technologies SIEM for Industries." The following individuals have my deepest gratitude for their invaluable advice, assistance, suggestions, and encouragement during my B.Sc. (Engg.) project.

Professor Dr. PARESH CHANDRA BARMAN, my supervisor, I am grateful for the time you gave me to complete my B.Sc. project. My B.Sc. project was completed with the aid of your suggestions, criticism, guidance, evaluation, and encouragement.

I want to thank all of my teachers for their guidance, assistance, inspiration, and encouragement during my time at the university.

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Chapter 1

Introduction

1.1 Background

The establishment of a Security Operations Center (SOC) has become a necessity for modern organizations due to the increasing scale and complexity of cyber threats. A SOC functions as the central hub of cybersecurity operations, designed to continuously monitor, detect, investigate, and respond to security incidents. Its role is to provide real-time visibility into network activities, ensuring timely identification and mitigation of potential risks.

Traditionally, developing a SOC has required heavy financial and technical investment, often relying on commercial or proprietary security solutions. Such high costs have made SOC implementation unattainable for many small and medium-sized enterprises (SMEs) and organizations with limited resources.

In this context, open-source Security Information and Event Management (SIEM) tools offer a practical alternative. SIEM systems form the backbone of SOC operations by collecting, correlating, and analyzing security events and log data from multiple sources, including servers, endpoints, applications, and network devices. By adopting open-source SIEM solutions, organizations can establish cost-effective SOCs that provide efficient monitoring, detection, and response to cyber threats without the constraints of vendor lock-in or excessive costs.

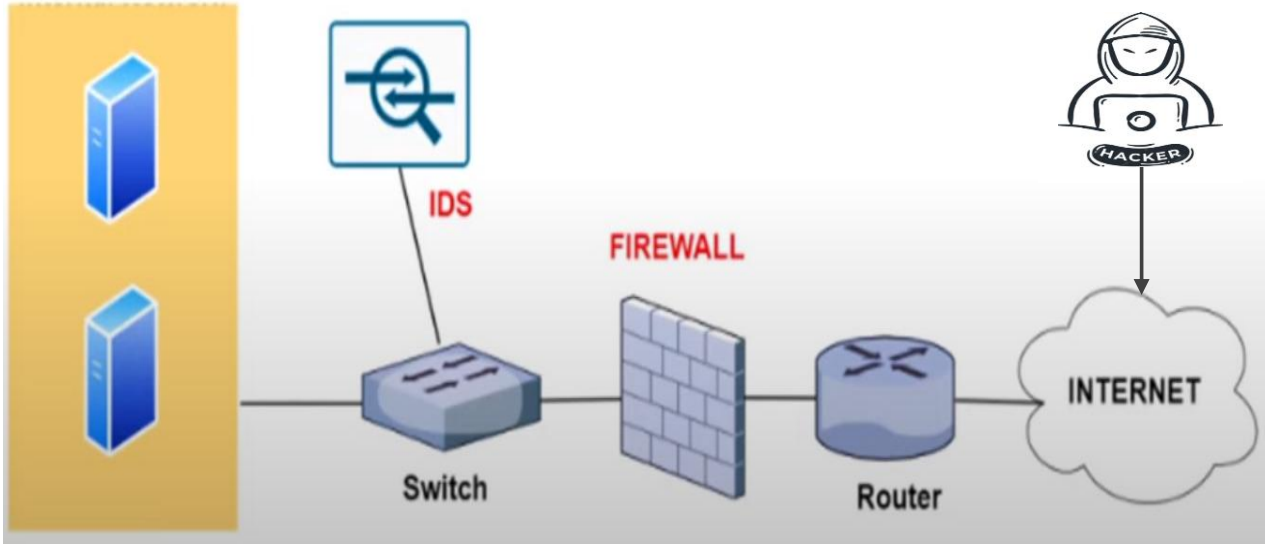


Figure 1. 1 Proposed SOC-Based SIEM Security System (source: https://www.youtube.com/watch?v=91i7InHVOso&t=10s)

As cybersecurity experts became aware of the necessity to comprehend new threats and modify defences accordingly, the development of SOC-based SIEM security solutions accelerated. The following significant elements are part of the history of these systems:

- Threat Intelligence
- Early Detection and Prevention
- Research and Collaboration

In conclusion, proactive defensive strategies, threat intelligence, early detection, and collaboration are at the core of SOC-based SIEM security systems. Organizations may improve their security posture, learn useful information about attackers, and take part in the global effort to combat cyber threats by strategically placing decoy systems.

1.2 Motivation

Industries today face growing cyber threats that demand continuous monitoring, rapid detection, and effective response. While traditional Security Operations Centers (SOC) are essential, their reliance on costly proprietary solutions limits adoption, especially for small and medium-sized organizations. Open-source SIEM technologies provide a practical alternative by offering affordability, flexibility, and transparency without compromising security.

This project is motivated by the need to design and implement a cost-effective SOC using open-source SIEM tools for industries. The solution aims to enhance threat detection, streamline incident response, and strengthen overall cybersecurity resilience. By leveraging open-source technologies, this work seeks to bridge the gap between advanced security requirements and financial constraints, enabling industries of all scales to operate in a safer digital environment..

1.3 Problem Statement

With the rapid digital transformation across industries, the risk of cyberattacks has grown significantly. Hackers use advanced tools and techniques such as port scanning, malware injection, credential theft, and denial-of-service attacks to exploit vulnerabilities in enterprise networks. Traditional security measures like firewalls and antivirus solutions are no longer sufficient to detect or respond to these sophisticated threats in real time.

Commercial Security Operations Center (SOC) solutions with integrated Security Information and Event Management (SIEM) are effective but often come with high costs, vendor lock-in, and limited customization. As a result, small and medium-sized industries struggle to deploy a comprehensive SOC for continuous monitoring, incident detection, and response.

Therefore, there is a critical need to design and implement a cost-effective SOC using open source technologies and SIEM that provides industries with real-time monitoring, log correlation, intrusion detection, and automated response ensuring security, scalability, and affordability without compromising performance.

More industries used Host-based Intrusion Detection Systems (HIDS). Host-based Intrusion Detection Systems (HIDS) provide deep visibility into system logs, file integrity, and processes, but they are limited to the specific host and cannot fully capture network-level attacks.

1.4 Probable Solution

We will design our system in such a way that when an attacker attacks, We can capture it from the on a switch or NIC card and monitor it in real time before it enters our system.We will use Suricata tool for monitoring and capture the network packets. Suricata operates in promiscuous mode, enabling its network interface card to capture and analyze all traffic passing through the switch, not just the packets addressed to it. This allowa suricata to detect suspicious activities such as intrusion attempts, port scans and malicious communications between the attacker machine, firewall and internal network. The generated alerts are then forwarded to the wazuh agent running on the suricata server. Wazuh agent collect the data and forward wazuh server. Wazuh server analysis that data then display on wazuh dashboard. Before, if we can be understand attacker try to enter our system, we can prevent the attacks.

1.5 Objective

The aim of this project is to design and implementation security operations center using open source technologies SIEM for industries. So, the objectives of this project are: (1) To learn more about SOC and open source SIEM solutions. (2) To obtain complete data security in an affordable manner and gain reliability for open source technology. (3) Gathering network device log events. (4) Detect harmful activity by analyzing logs and events

For deployment of the design and implementation security operations center using open source technologies SIEM for industries we used open-source tools. And they are-

Kali Linux: Popular Linux distribution Kali Linux was developed primarily for ethical hacking, digital forensics, and penetration testing. It is a strong and complete operating system that offers a wide range of tools and services for security. Based on the Debian distribution, Kali Linux provides a pre-configured environment with a number of tools and utilities for different security testing and analysis tasks. There are services for vulnerability evaluation, network scanning, password cracking, wireless attacks, web application testing, and more. Kali Linux is used by penetration testers, security specialists, and cybersecurity enthusiasts.

Ubuntu Desktop: Popular Linux-based operating system for desktops and laptops, Ubuntu Desktop. The goal of this user-friendly, adaptable distribution is to offer a free and open-source alternative to proprietary operating systems like Windows and macOS. Based on the Debian operating system, Ubuntu Desktop is created by Canonical Ltd.

Wazuh: An open-source security platform called Wazuh offers current IT infrastructures threat detection, visibility, and compliance capabilities. It creates a comprehensive security monitoring solution by fusing components of intrusion detection, log analysis, file integrity monitoring, and vulnerability evaluation.

Suricata: Suricata is open-source network intrusion detection and prevention system (IDS/IPS). It is intended to monitor network traffic and detect a variety of threats, including malware, intrusion attempts, and network anomalies. Using a rule-based language, Suricata analyzes network packets in real time, allowing it to identify and respond to suspicious or malicious

activities. The non-profit organization OISF (Open Information Security Foundation) owns and develops Suricata

Hydra: An effective and well-liked open-source program for brute-forcing passwords is called Hydra. It is intended to assist security experts in evaluating the security of authentication systems and password strength.

Nmap: Prominent open-source network scanning software called Nmap (Network Mapper) is used for network research and security audits. On a network, it is intended to find hosts, services, and vulnerabilities.

VMware Workstation: The desktop virtualization program VMware Workstation was created by VMware Inc. On a single physical computer, users may build and run several virtual machines (VMs). On a host machine, these virtual machines may run many operating systems simultaneously, including Windows, Linux, and macOS.

Security Onion: An open-source cybersecurity platform called Security Onion offers a number of tools for log management, intrusion detection, and network security monitoring. In network contexts, it is intended to make it easier to notice, analyze, and respond to security events.

1.6 Hypothesis

This project assumes that the design and implementation of a Security Operations Center (SOC) using open-source technologies and SIEM will enhance the detection and response capabilities of industrial networks. By deploying Suricata in promiscuous mode, it is hypothesized that malicious activities such as port scans, brute-force attempts, and DDoS attacks can be effectively captured and analyzed in real time. Wazuh, when integrated with Suricata, is expected to correlate alerts and present them in a centralized dashboard for efficient monitoring. The hypothesis also assumes that open-source tools, despite being free, can provide a reliable and scalable solution for threat detection when properly configured. Furthermore, it is believed that custom rules in Suricata will improve detection accuracy and reduce false positives. The integration of HIDS and NIDS approaches is expected to deliver a more holistic view of network and host activities. It is also hypothesized that such a SOC model can be adapted for industrial use without high infrastructure costs. Another assumption is that this system will improve incident response time through centralized visibility. Finally, the hypothesis posits that the proposed SOC framework can serve as a cost-effective alternative to expensive commercial SIEM platforms while ensuring adequate protection for industries.

Chapter 2

Literature Review

In recent years, the rapid increase in cyber threats directed at industrial networks ranging from manufacturing to critical infrastructure has driven interest toward more accessible, flexible, and cost-effective Security Operations Center (SOC) architectures. Open-source technologies have become especially attractive in this context, enabling industries with limited budgets to implement robust detection, response, and compliance capabilities without reliance on expensive proprietary software. Several recent research papers and real-world projects explore this trend, dissecting both benefits and challenges of SOC implementation using open-source SIEM technologies.

[1] Bassey, Chinda, & Idowu (2024) presented a "Building a Scalable Security Operations Center: A Focus on Open-Source Tools," in which they constructed a fully functional SOC architecture combining Wazuh as SIEM/XDR, Suricata for network intrusion detection, and TheHive for case management. Their experimental setup included simulation of brute force, malware download, and DoS attacks on endpoints. The results showed that the open-source stack could accurately detect and respond to these attacks, achieving good visibility into endpoint and network behavior. Importantly, they demonstrated that even with constrained computing resources, such a system could scale and remain effective in small to medium industrial environments.

Another relevant study is [2] "Hybrid Intrusion Detection System and Network Infrastructure Vulnerability Mitigation using Active Response (XDR) Technique Wazuh and Suricata," by Damanik & Anggraeni. This work focused on practical deployment in organizational network infrastructure, integrating Wazuh agents with Suricata and using Wazuh's active response features. They tested reconnaissance, brute-force, and DoS attacks and showed that malicious IPs could be identified and actively blocked. This highlights the value of integrating detection (Suricata) with response (Wazuh's active response) in an industrial context.

Privacy, regulatory compliance, and heterogeneous network environments are other major focus areas in recent literature. A study [3] "Implementing and evaluating a GDPR-compliant open-source SIEM solution" (2023) examined how open-source SIEMs must be designed and configured to meet privacy regulations while ingesting logs from many types of devices and operating systems. It emphasizes structured log collection, proper anonymization/pseudonymization, and role-based access controls as essential for industrial deployment.

More advanced work is emerging that enhances SOC capability beyond signature or rule-based detection. For example, [4] "Enhancing Security Operations Center: Wazuh Security Event Response with Retrieval-Augmented-Generation-Driven Copilot" (Ismail et al., 2025) integrates AI with Wazuh to assist analysts. Here, Wazuh is used to aggregate and correlate events, and then an AI "copilot" leverages MITRE ATT&CK and NIST CSF to provide contextually relevant response suggestions. This addresses a common gap: the backlog in incident response due to high alert volume and limited analyst capacity. The study showed that this hybrid approach can

significantly improve response times and decision accuracy in industrial-like scenarios.

Another relevant angle is the work on SIEM evasion. [5] “You Cannot Escape Me: Detecting Evasions of SIEM Rules in Enterprise Networks” (2023) exposes that many enterprise SIEM rule sets have blind spots: adversaries can evade a large percentage of them using common techniques. The authors propose AMIDES, an adaptive misuse detection system layered over standard SIEM rule sets, to catch those evasions. For SOCs in industrial settings, this suggests that relying solely on predefined or rule-based detection (even if open source) may be insufficient; adaptive and context-aware detection layers are becoming essential.

Chapter 3

Design & Implementation

3.1 Design

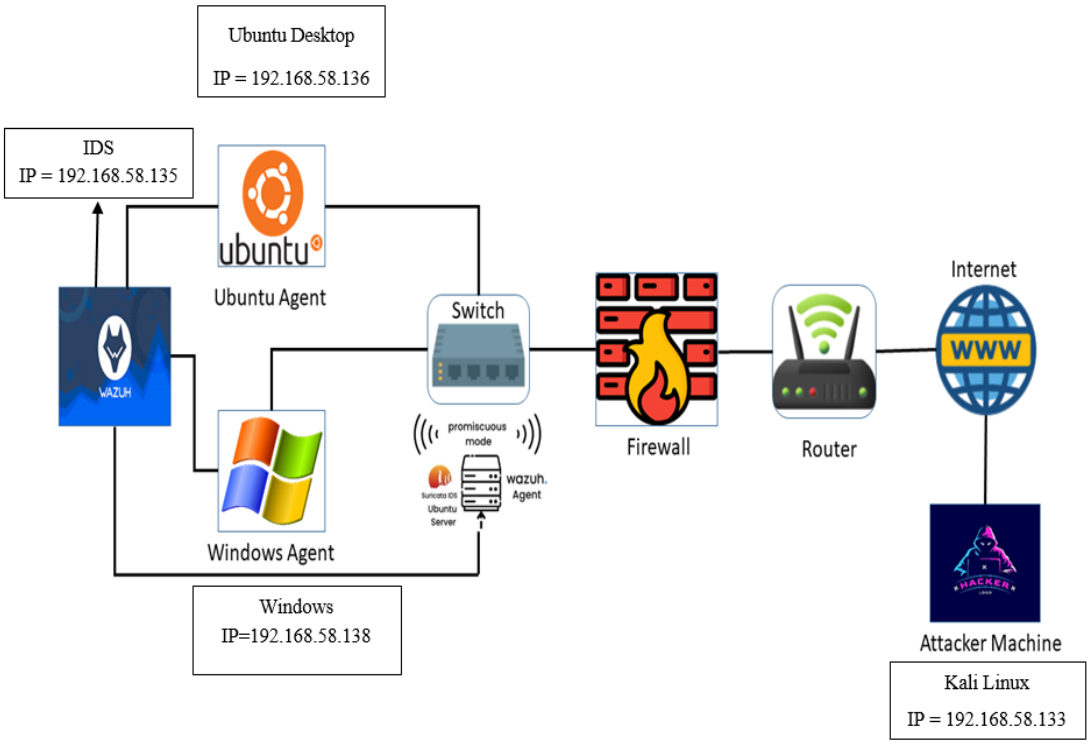


Figure 3.1 Implemented Design and Implementation Security Operations Center (SOC) Using Open Source Technologies SIEM for Industries

Explanation of Figure 3.1

The diagram is the practical design of a Security Operations Center (SOC) built with open-source SIEM tools. It demonstrates how different components of the network interact to detect and respond to cyber threats.

- The attacker machine (Kali Linux, IP 192.168.58.133) represents an external threat coming from the internet.
- The router and firewall control traffic entering the internal network, providing the first layer of defence.
- Inside the network, a switch or NIC operating in promiscuous mode forwards traffic to SOC monitoring system.
- Wazuh agent are installed on both Ubuntu (IP 192.168.58.136) and Windows (IP 192.168.58.138) systems to collect security logs and send them to the wazuh server/IDS

(IP 192.168.58.135).

- The IDS inspects traffic and generates alerts when suspicious activities are detected
- This setup allows analysts to monitor, detect and respond to malicious behavior in real time, ensuring stronger protection for the network

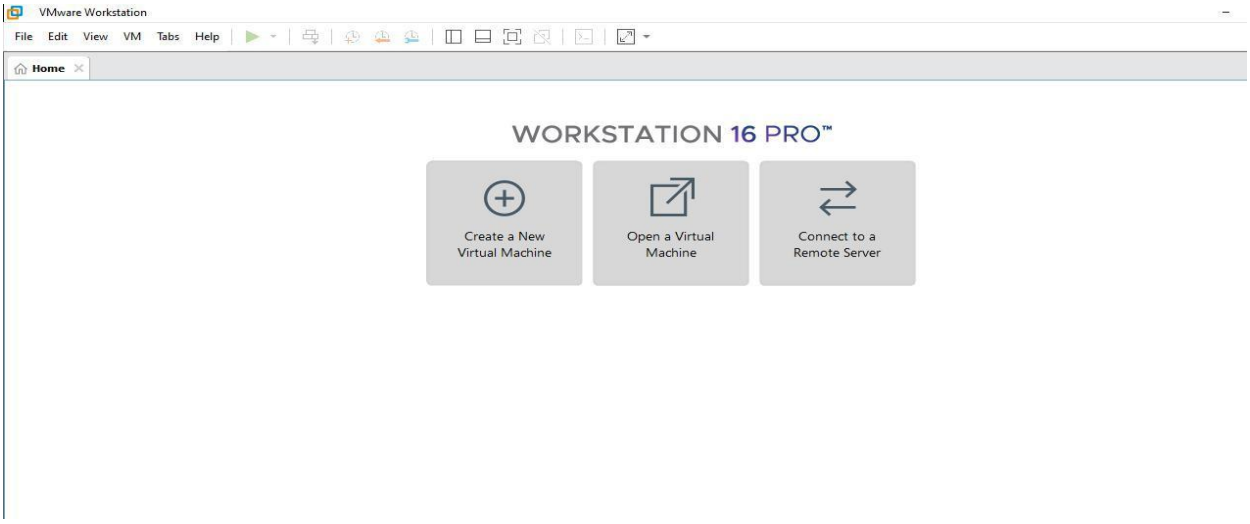
3.2 Implementation

3.2.1 Install VMware Workstation

The popular virtualization program VMware Workstation was created by VMware, Inc. On a single physical computer, users may build and run several virtual machines (VMs). You can build virtual machines using VMware Workstation that replicate the CPU, memory, storage, and network adapters of a real computer. This eliminates the need for separate physical devices so that you may run several operating systems like Windows, Linux, and macOS concurrently on your computer. Here we download the VMware Workstation from-

https://www.vmware.com/products/workstation-pro.html

And we install it on our Windows host.



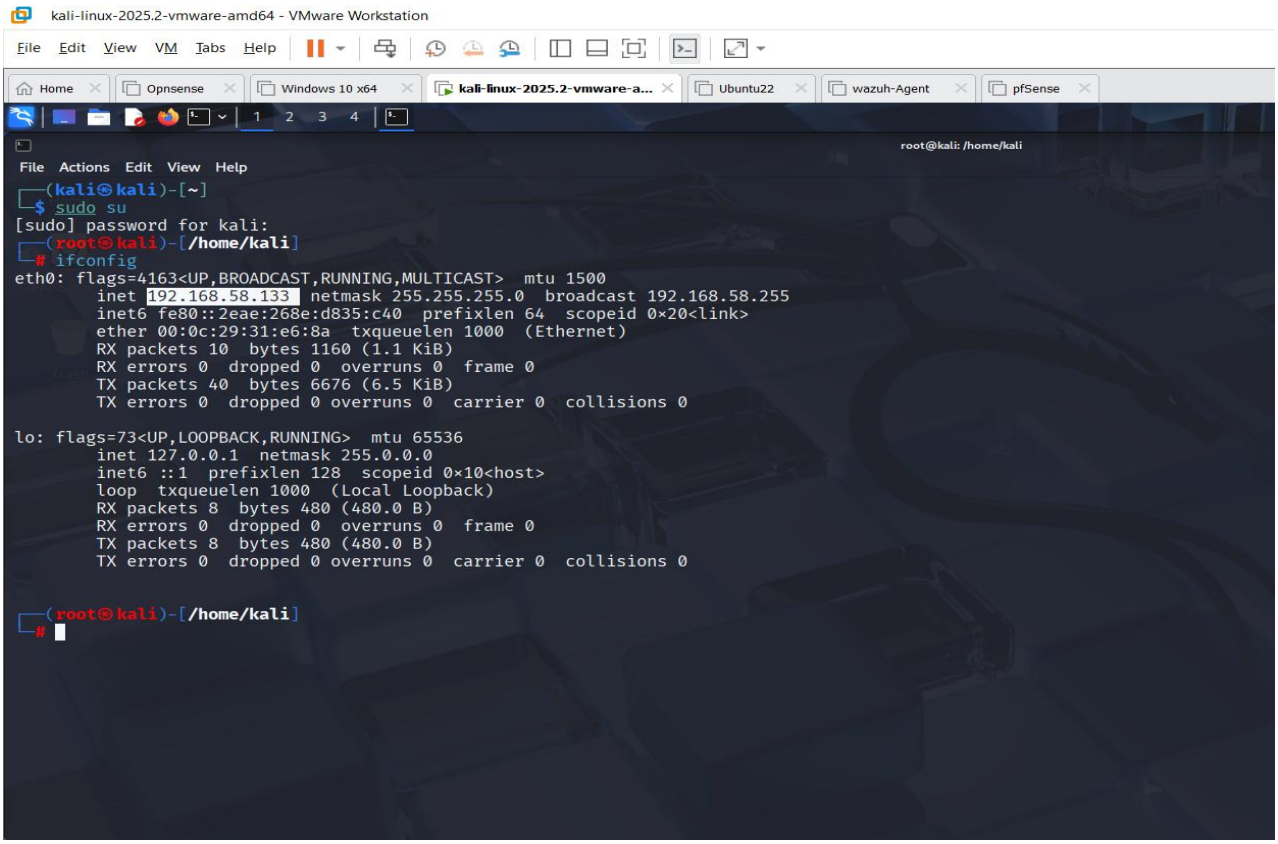
3.2.2 Install & Configure Kali Linux

A specific Linux distribution called Kali Linux is made for network security evaluations, ethical hacking, and penetration testing. Leading cybersecurity training and penetration testing business Offensive Security created and maintains it. So we choose the kali Linux as an attacker machine. Here we download the Kali Linux from-

https://www.kali.org/get-kali/#kali-platforms

And we install it on VMware Workstation. For Displaying the IP address we use the command-

ifconfig



Here,

Username: kali, root

IP Address: 192.168.58.133

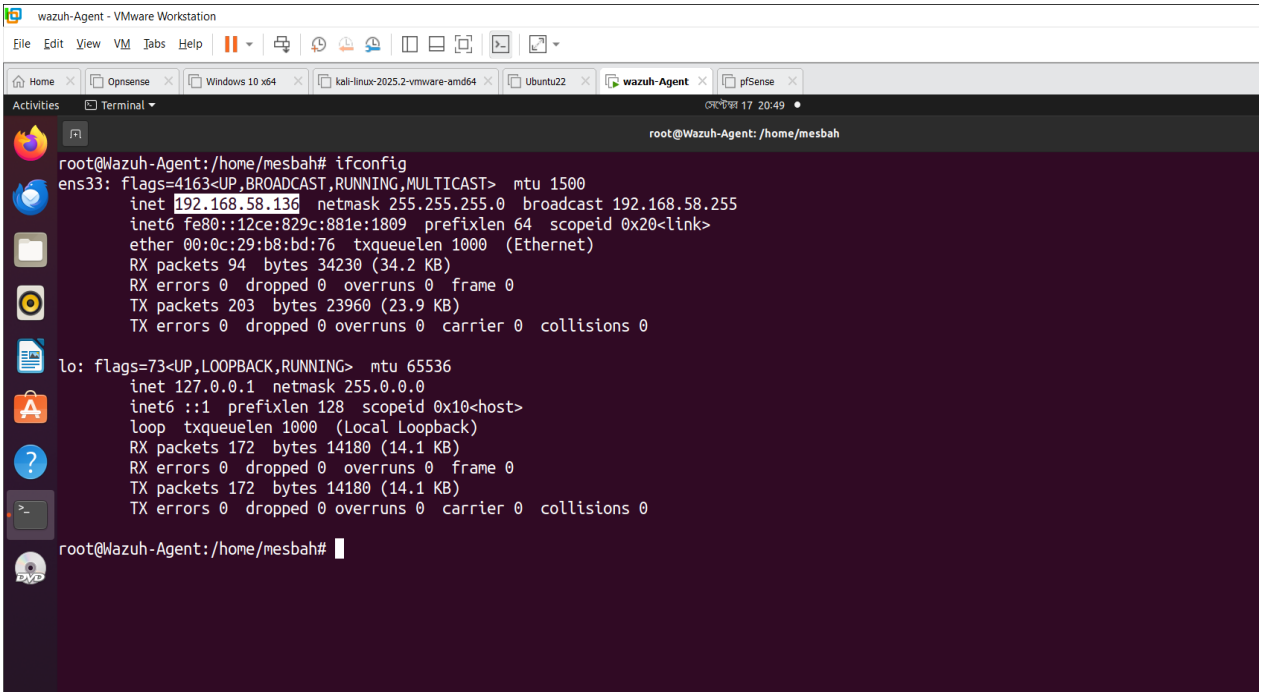
3.2.3 Install & Configure Ubuntu Desktop

An open-source, Linux-based operating system called Ubuntu Desktop was created by Canonical Ltd. One of the most extensively used and well-known Linux distributions, it is renowned for its simple to use interface. Here we used Ubuntu desktop as a Wazuh agent. The Ubuntu desktop Download from-

https://ubuntu.com/download/desktop

And we install it on VMware Workstation. For displaying the IP address we use the command-

ifconfig



Here,

Username: Wazuh-Agent, root

IP Address: 192.168.58.136

3.2.4 Install Suricata IDS Package & Configure

Here we install suricata version : 7.0
we install Suricata on the Wazuh Agent. We use the command-

sudo add-apt-repository ppa:osif/suricata-stable
sudo apt-get update
sudo apt-get install suricata -y

Download and extract the Emerging Threats Suricata ruleset:

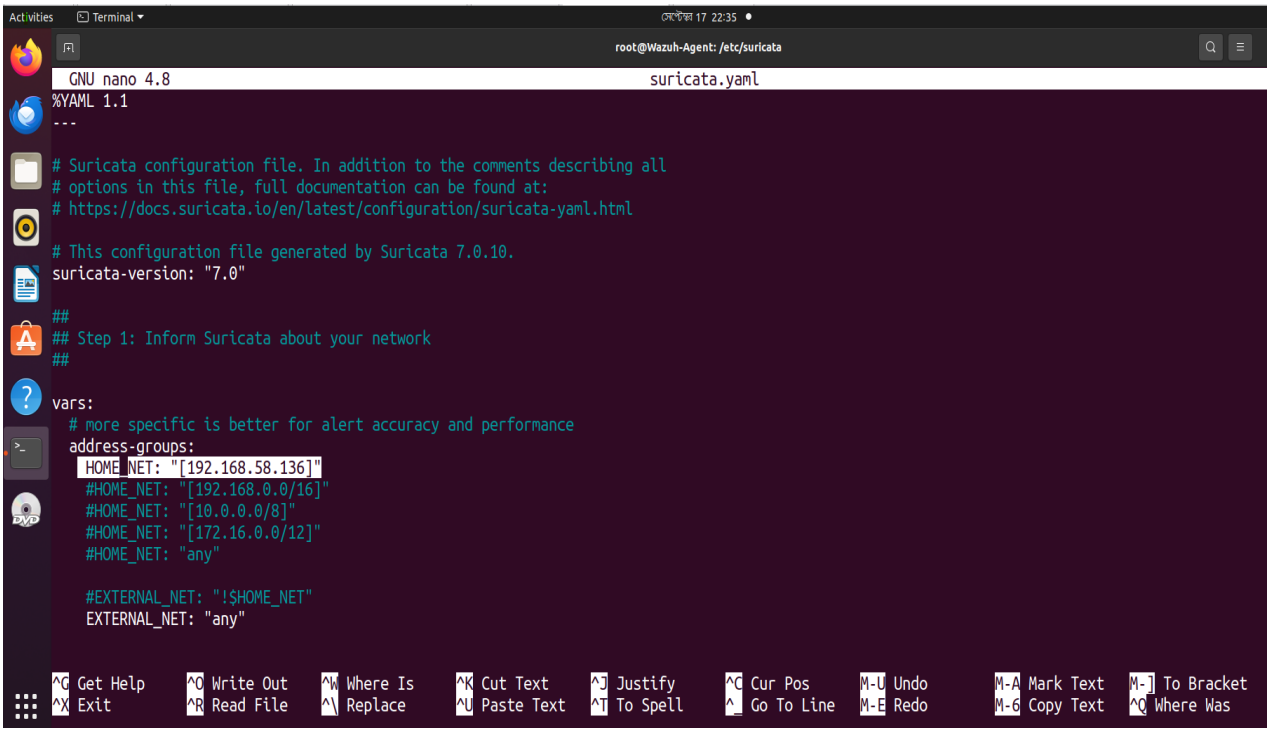
cd /tmp/ && curl –LO https://rules.emergingthreats.net/open/suricata-6.0.8/emerging.rules.tar.gz
sudo tar –xvzf emerging.rules.tar.gz && sudo mv rules/*.rules /etc/suricata/rules/
sudo chmod 640 /etc/suricata/rules/*.rules

Modify Suricata setting in the /etc/suricata/suricata.yaml file:

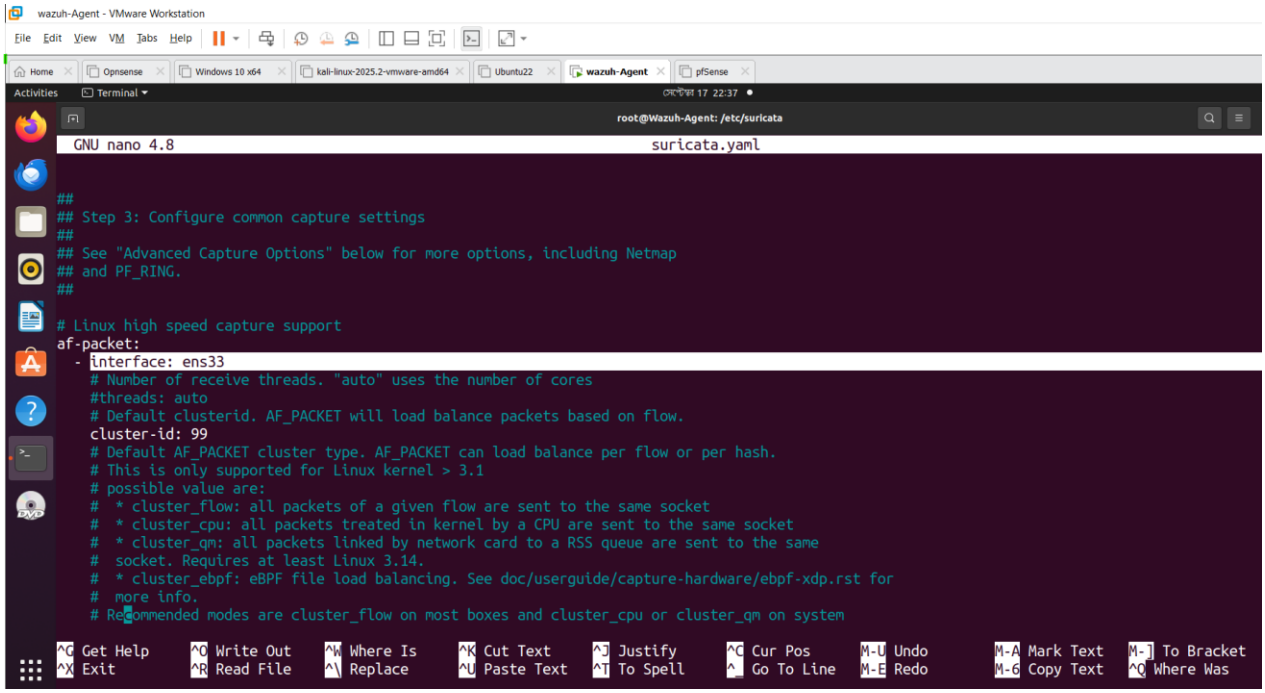
```
sudo nano /etc/suricata/suricata.yaml
```

First setup HOME_NET: 192.168.58.136

Secondly setup EXTERNAL_NET: any



Finally setup the NIC interface: ens33



Restart the Suricata use this command-

```
sudo systemctl restart suricata
```

Suricata integrate with Wazuh:

```
sudo nano /var/ossec/etc/ossec.conf
```

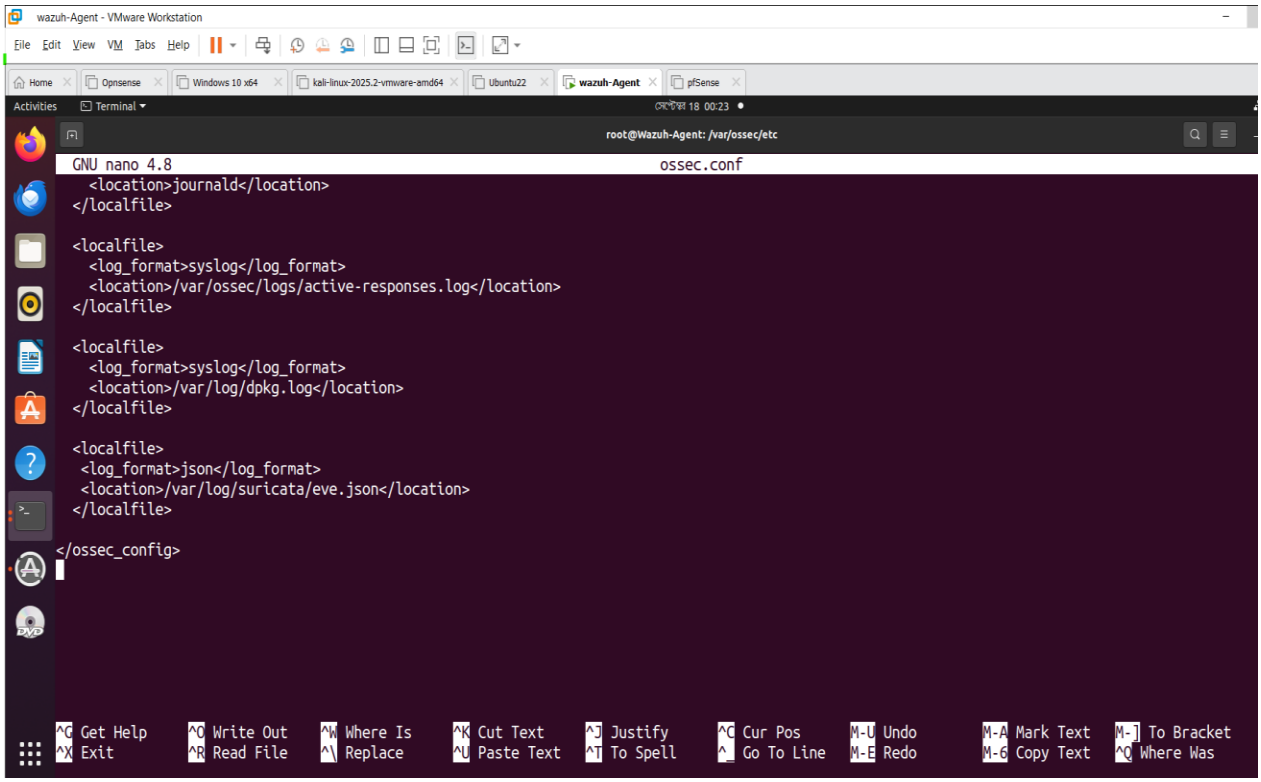
Add the suricata json path on ossec.conf :

<localfile>

<log_format>json</log_format>

<location>/var/log/suricata/eve.json</location>

</localfile>



3.2.5 Install & Configure Windows 10

The most recent major version of the Windows operating system is Windows 10. It is an extremely common and well-liked operating system for desktops, laptops, and tablets. Here we install windows 10 machine as a wazuh agent. We Install the Windows 10 machine on VMware

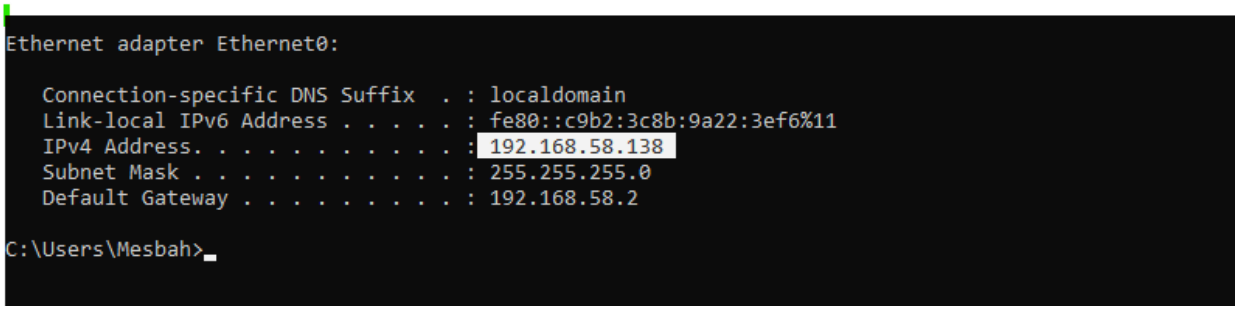
Workstation.

Windows 10 operating system was downloaded from-

https://www.microsoft.com/en-us/software-download/windows10

For Displaying the IP Address, The command is-

```
ipconfig
```



Here,

Username: Mesbah

IP Address: 192.168.58.138

3.2.6 Install & Configure Wazuh

Open Virtual Appliance (OVA) format pre-built virtual machine images are made available by Wazuh. This is immediately importable into VirtualBox or other OVA-compliant virtualization programs. Remember that this VM only functions on 64-bit operating systems. It does not come standard with high availability and scalability. However, distributed deployment may be used to implement this.

We download the Wazuh OVA file from-

https://documentation.wazuh.com/current/deployment-options/virtual-machine/virtual-machine.html

And we install or deploy the wazuh.ova file on VMware Workstation. It was a Cent-OS-based server. For displaying the IP address-

ip add

```
[root@wazuh-server ~]# ip add
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 100
0
    link/ether 00:0c:29:b0:02:39 brd ff:ff:ff:ff:ff:ff
    inet 192.168.225.138/24 brd 192.168.225.255 scope global dynamic eth0
       valid_lft 1497sec preferred_lft 1497sec
    inet6 fe80::20c:29ff:feb0:239/64 scope link
       valid_lft forever preferred_lft forever
[root@wazuh-server ~]#
```

Here,

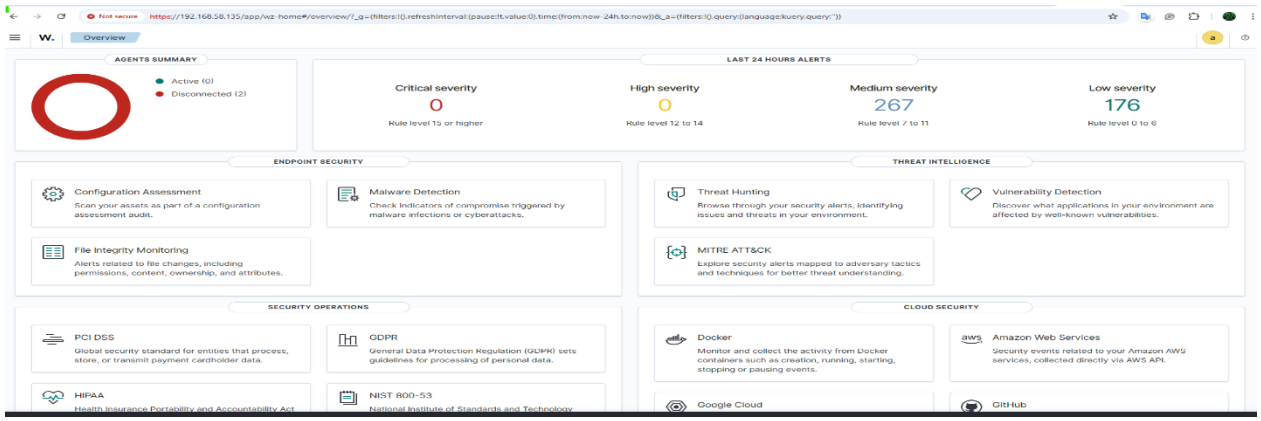
Username: root

Password: wazuh

IP Address: 192.168.225.138

We used several commands for starting the wazuh manager, wazuh dashboard, wazuh agent and deploy the wazuh API.

systemctl start wazuh-manager
systemctl start wazuh-dashboard
systemctl start wazuh-agent



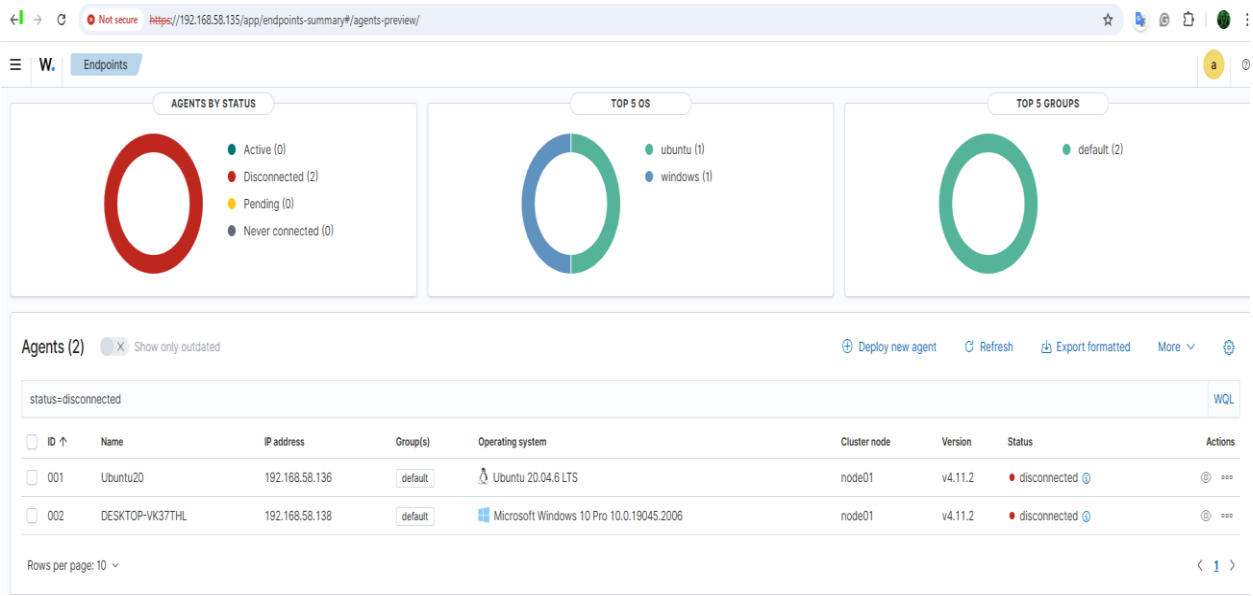
3.2.7 Adding Agent on Wazuh

For adding the windows machine with wazuh here we used the command-

Invoke-WebRequest -Uri https://packages.wazuh.com/4.x/windows/wazuh-agent-4.4.3-1.msi -OutFile ${env:tmp}\wazuh-agent.msi; msiexec.exe /i ${env:tmp}\wazuh-agent.msi /q WAZUH_MANAGER='192.168.58.135' WAZUH_REGISTRATION_SERVER='192.168.58.135' WAZUH_AGENT_GROUP='default' WAZUH_AGENT_NAME='windows_main_system'
NET START WazuhSvc

For adding the Ubuntu Desktop , the command were used-

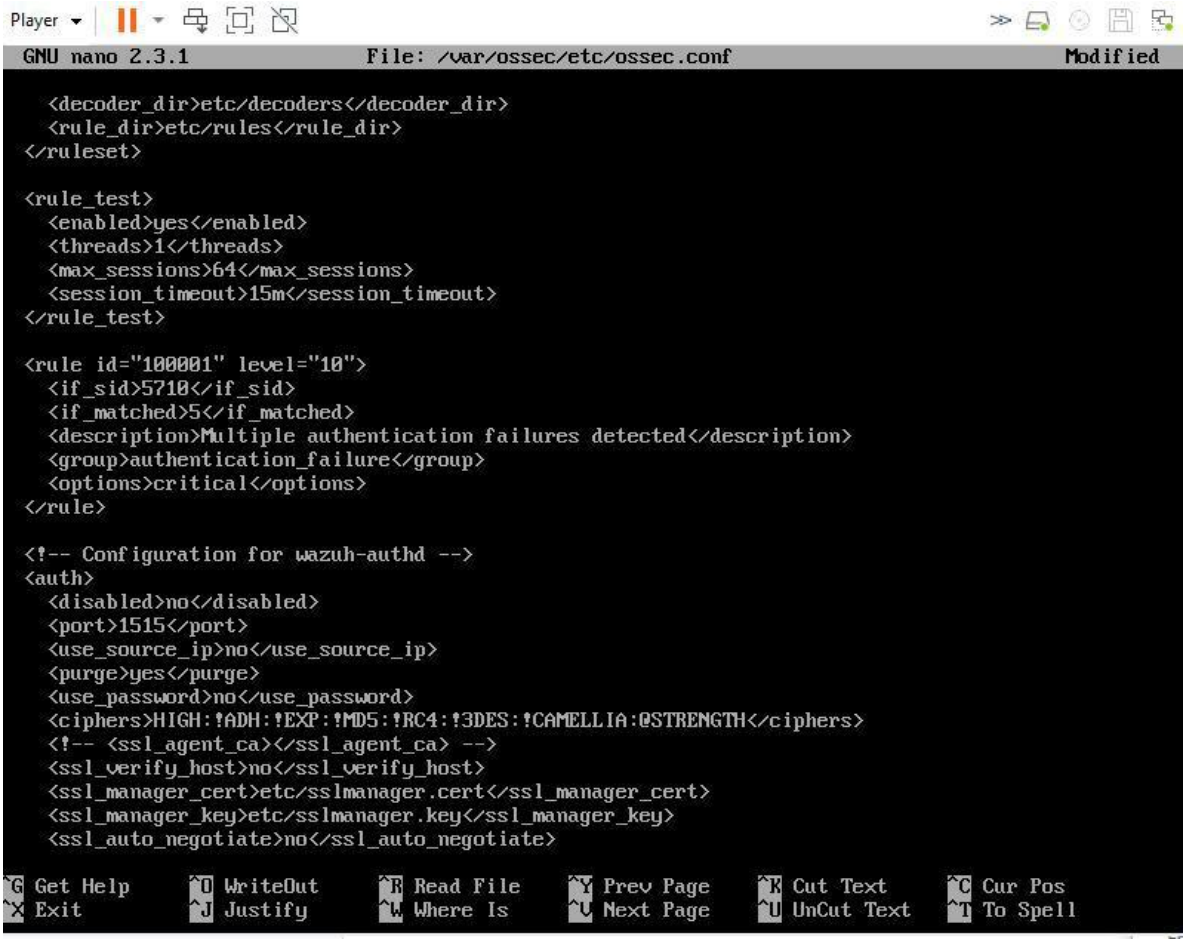
curl -so wazuh-agent.deb https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.4.3-1_amd64.deb && sudo WAZUH_MANAGER='192.168.58.135' WAZUH_AGENT_GROUP='default' WAZUH_AGENT_NAME='Ubuntu_Agent' dpkg - i ./wazuh-agent.deb
sudo systemctl daemon-reload
sudo systemctl enable wazuh-agent
sudo systemctl start wazuh-agent



3.2.8 Set Rule on Wazuh

Here I set a rule for better detecting the Broken authentication attack on the Wazuh configuration script. And the script was-

```
<rule id="100000" level="10">
  <if_sid>5715</if_sid> <!-- The SID of the authentication failure event -->
  <match> <!-- Specify the conditions for triggering the rule -->
    <threshold field="srcip" count="5" time="60"> <!-- 5 times in 60 seconds -->
      <groupby>srcip</groupby>
    </threshold>
  </match>
  <description>Multiple authentication failures from a single IP address.</description>
</rule>
```



Here wazuh detect a session alert level 10 when happening failure authentication more than 5 times.

Chapter 4

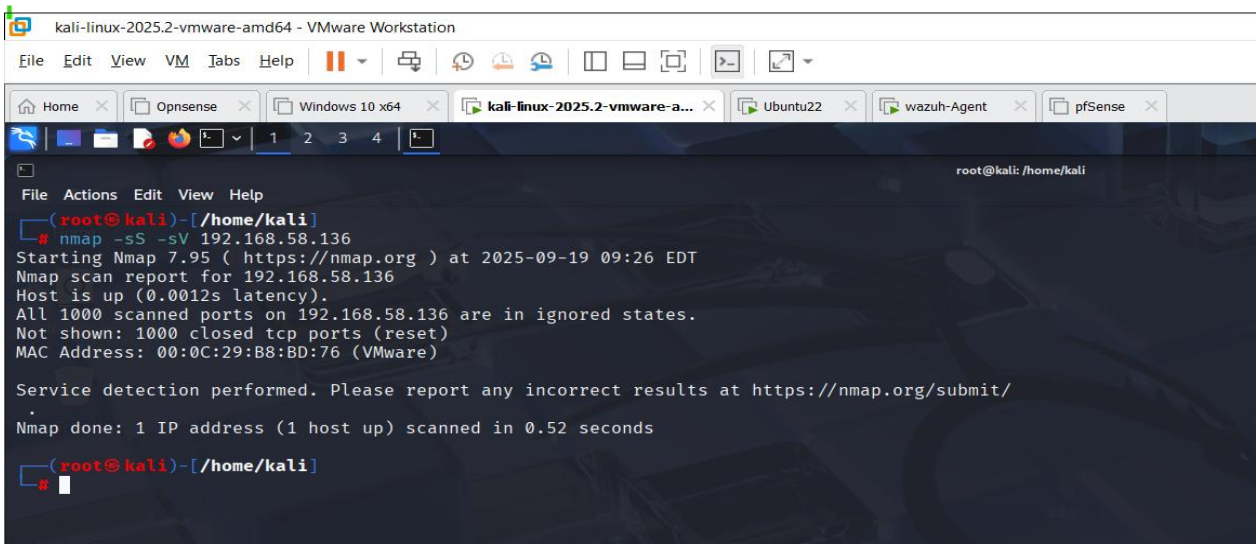
Launching Attacks

4.1 Enumeration

Enumeration is a cybersecurity method for learning more about a target network or system. It comprises carefully approaching a target and probing it to find out about its capabilities, settings, and services. Both manual and automatic enumeration are possible with the use of tools and software.

For the finding of open port of the Target machine, we scan the all port. And we use Nmap tools. Which is actually do Network Mapping.

```
nmap -sS -sV 192.168.58.136
```



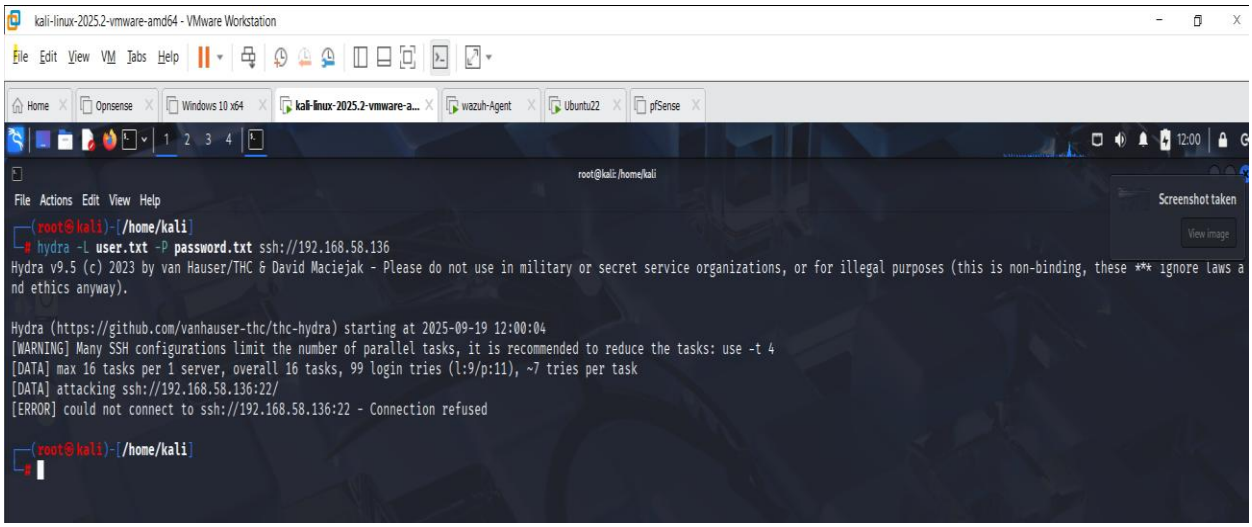
Here, we perform the scanning on our target machine. Attempt to identify any open ports on the target machine.

4.2 Brute Force Attack

A brute force attack is a sort of cybersecurity attack in which an attacker carefully tries all possible password or encryption key combinations until the correct one is discovered. A brute force assault seeks illegal access to a system, user account, or encrypted data. With the base of Enumeration, Here I conduct the brute force attack with Hydra which is a very popular brute force tool. For Collecting the user and password for SSH connection, I conducted the brute force attack.

The command for performing the attack-

```
hydra -L user.txt -P password.txt ssh://192.168.58.136
```



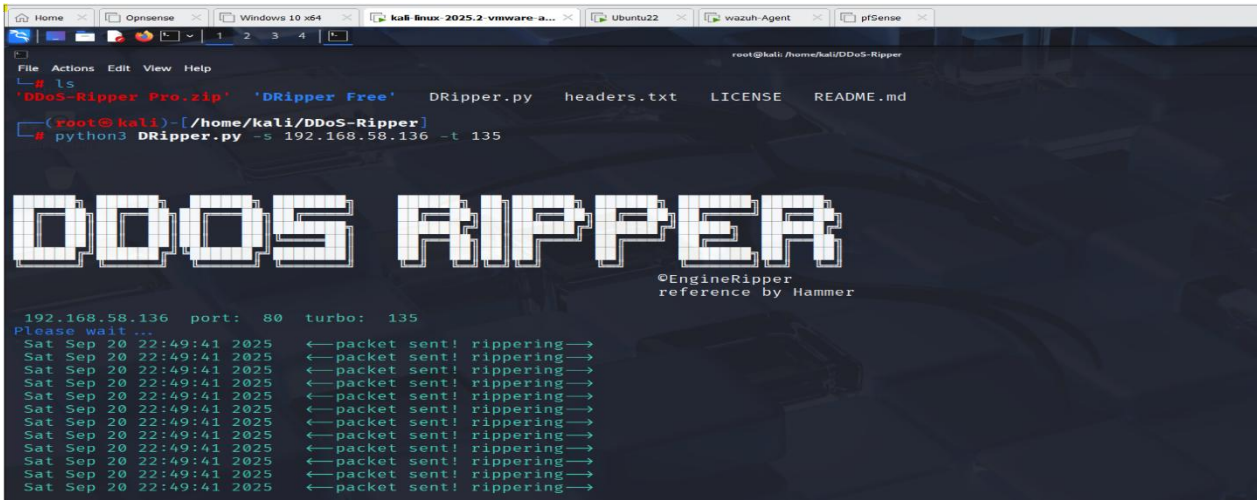
Here my target IP was 192.168.58.136. Actually I perform the brute force attack with two wordlists and that were, User.txt and Password.txt.

4.3 DDOS attack

A distributed denial of service (DDoS) attack is a malicious attempt to stop a network, service, or website from operating normally by saturating it with unwelcome traffic. The goal of a DDoS assault is to exhaust the target's resources, such as bandwidth, processing power, or memory, rendering it inaccessible to authorized users.

I perform the DDOS attack with DDOS ripper. And the commands was-

https://github.com/palahsu/DDoS-Ripper.git
python3 DRipper.py -s 192.168.58.136 –p 22 -t 135



Here, We can see that a large number of request is offered from the attacking machine to the victim machine. During the attack the victim machine network remain down.

Chapter 5

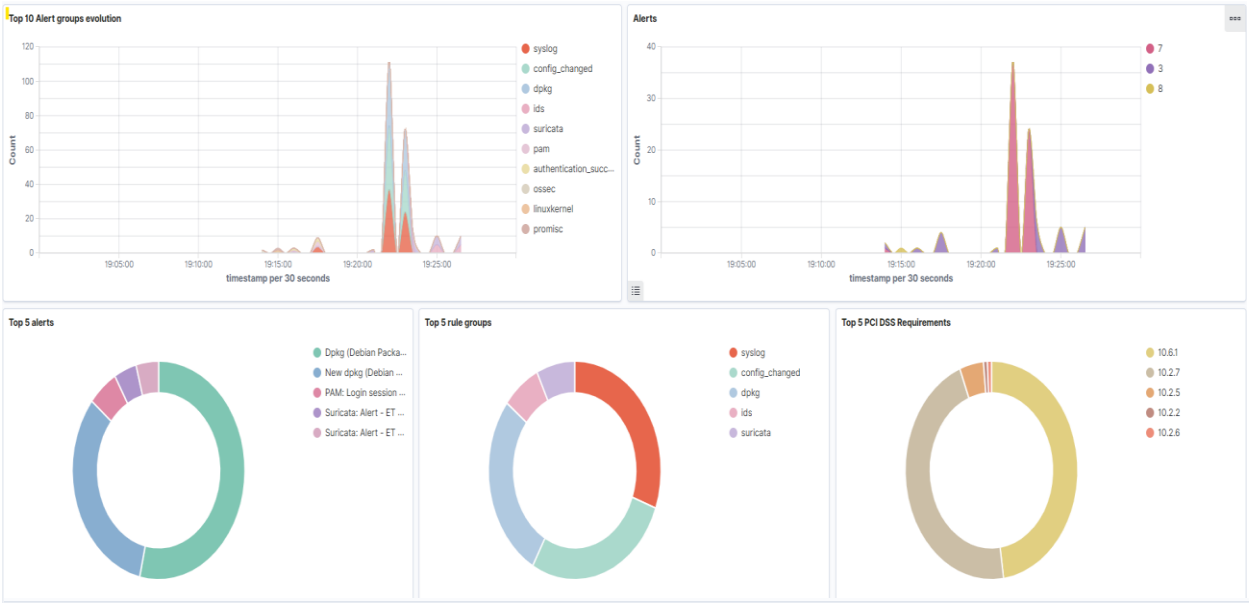
Threat Detection & Analysis

Here we used IDS & SIEM Technology for Detecting the Attack. Network packets or system logs are analyzed by an IDS either in real-time or after the fact. To find patterns or signs of potential security breaches or assaults, it evaluates the observed activity against predetermined rules, signatures, or anomaly detection algorithms. The NIDS can provide alarms, log events, or execute automated measures to reduce the danger when suspicious behavior is found. we used Suricata and Wazuh which has Intrusion Detection System. Which is also known as SIEM Technology. All attacks that we perform before all are detected by the Suricata NIDS or Wazuh.

5.1 Detecting Enumeration

During the Enumeration, we generally gather information using different tools. In previous, we perform an enumeration using Nmap or Network Mapper tools. Our PC has 65535 ports. Nmap establishes a full TCP connection with the target port to determine its open or closed state.

The Suricata NIDS also detect here,



616 hits

Sep 18, 2025 @ 19:26:50.570 - Sep 19, 2025 @ 19:26:50.570

Export Formatted | 733 available fields | Columns | Density | 1 fields sorted | Full screen

timestamp	agent.name	rule.description	rule.level	rule.id
Sep 19, 2025 @ 19:26:41.1...	Ubuntu20	Suricata: Alert - ET SCAN Potential VNC Scan 5800-5820	3	86601
Sep 19, 2025 @ 19:26:41.1...	Ubuntu20	Suricata: Alert - ET SCAN Suspicious inbound to PostgreSQL port 5432	3	86601
Sep 19, 2025 @ 19:26:41.1...	Ubuntu20	Suricata: Alert - ET SCAN Suspicious inbound to Oracle SQL port 1521	3	86601
Sep 19, 2025 @ 19:26:41.1...	Ubuntu20	Suricata: Alert - ET SCAN Suspicious inbound to MSSQL port 1433	3	86601
Sep 19, 2025 @ 19:26:41.0...	Ubuntu20	Suricata: Alert - ET SCAN Suspicious inbound to mySQL port 3306	3	86601
Sep 19, 2025 @ 19:25:02.5...	Ubuntu20	Suricata: Alert - ET SCAN Suspicious inbound to PostgreSQL port 5432	3	86601
Sep 19, 2025 @ 19:25:02.5...	Ubuntu20	Suricata: Alert - ET SCAN Suspicious inbound to mySQL port 3306	3	86601
Sep 19, 2025 @ 19:25:02.5...	Ubuntu20	Suricata: Alert - ET SCAN Suspicious inbound to Oracle SQL port 1521	3	86601

Suricata was configured in promiscuous mode to monitor the interface *ens33* on the Ubuntu 20 Wazuh agent. When an attacker scanned the victim machine using Nmap, Suricata captured the packets from the *ens33* interface and analyzed them using Suricata rules. Upon detection, Suricata generated an alert and sent it to the Wazuh server, where the alert was displayed on the Wazuh dashboard.

5.2 Detecting Brute Force Attack

A brute force attack is an effort to obtain unauthorized access to a target system or account by repeatedly testing all login and password combinations until the right ones are discovered. So, there happening a large number of authentication failure.

Document Details

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data.event_type	alert
data.flow.bytes_toclient	0
data.flow.bytes_toserver	74
data.flow.dest_ip	192.168.58.136
data.flow.dest_port	22
data.flow.pkts_toclient	0
data.flow.pkts_toserver	1
data.flow.src_ip	192.168.58.133
data.flow.src_port	38278
data.flow.start	2025-09-21T14:01:59.891987+0600
data.flow_id	2142208361159842.000000
data.in_iface	ens33
data.pkt_src	wire/pcap
data.proto	TCP
data.src_ip	192.168.58.133
data.src_port	38278
data.timestamp	Sep 21, 2025 @ 14:01:59.891
decoder.name	json
id	1758441720.867201
input.type	log
location	/var/log/suricata/eve.json
manager.name	wazuh-server
rule.description	Suricata: Alert - BRUTE_FORCE HYDRA - SSH SYN flood/bruteforce
rule.firedtimes	4
rule.groups	ids, suricata
rule.id	86601

Here , Suricata detect:

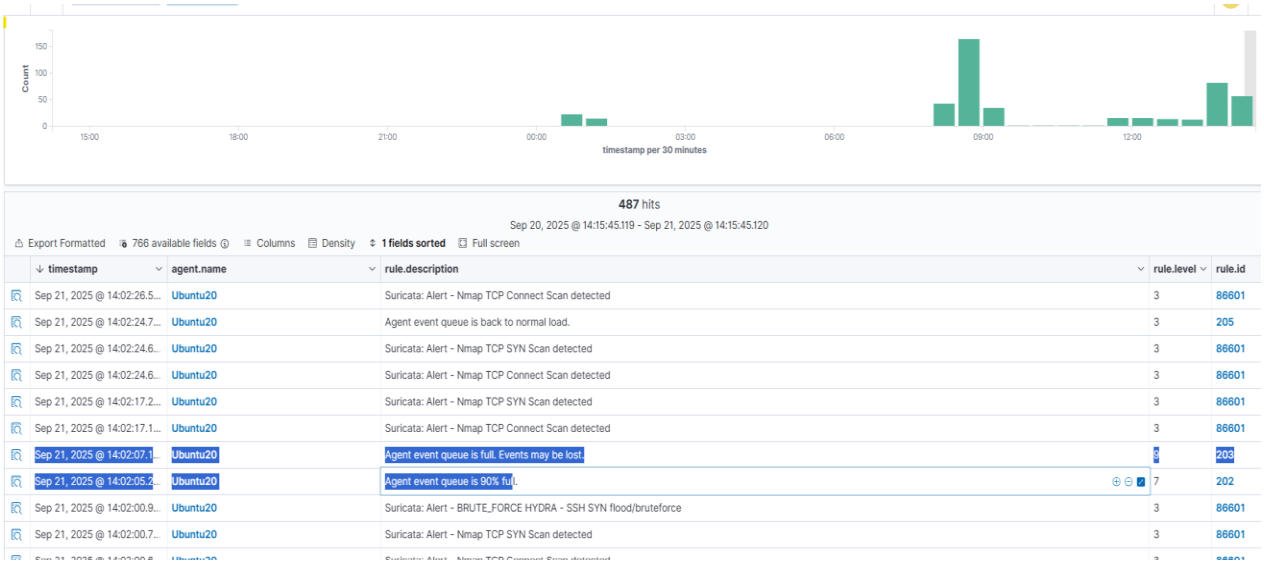
Suricata Alert – BRUTE_FORCE HYDRA – SSH SYN flood/bruteforce

Data source IP = 192.168.58.133

Data destination IP = 192.168.58.136

5.3 Detecting DDOS Attack

A DDoS (Distributed Denial-of-Service) attack is a cyberattack that floods a targeted server, service, or network with a massive amount of internet traffic, making it unavailable to legitimate users. These attacks use multiple compromised computer systems, often in a botnet, to generate the overwhelming traffic, which can be more effective and harder to detect than an attack from a single source. The goal is to disrupt operations, leading to poor performance or a complete shutdown of the target.



timestamp	agent.name	rule.description	rule.level	rule.id
Sep 21, 2025 @ 14:02:26.5...	Ubuntu20	Suricata: Alert - Nmap TCP Connect Scan detected	3	86601
Sep 21, 2025 @ 14:02:24.7...	Ubuntu20	Agent event queue is back to normal load.	3	205
Sep 21, 2025 @ 14:02:24.6...	Ubuntu20	Suricata: Alert - Nmap TCP SYN Scan detected	3	86601
Sep 21, 2025 @ 14:02:24.6...	Ubuntu20	Suricata: Alert - Nmap TCP Connect Scan detected	3	86601
Sep 21, 2025 @ 14:02:17.2...	Ubuntu20	Suricata: Alert - Nmap TCP SYN Scan detected	3	86601
Sep 21, 2025 @ 14:02:17.1...	Ubuntu20	Suricata: Alert - Nmap TCP Connect Scan detected	3	86601
Sep 21, 2025 @ 14:02:07.1...	Ubuntu20	Agent event queue is full. Events may be lost.	9	203
Sep 21, 2025 @ 14:02:05.2...	Ubuntu20	Agent event queue is 90% full.	7	202
Sep 21, 2025 @ 14:02:00.9...	Ubuntu20	Suricata: Alert - BRUTE_FORCE HYDRA - SSH SYN flood/bruteforce	3	86601
Sep 21, 2025 @ 14:02:00.7...	Ubuntu20	Suricata: Alert - Nmap TCP SYN Scan detected	3	86601

Here, Suricata detect:

Ubuntu20 wazuh agent , Agent event queue is 90% full then finally agent event queue is full. Events may be lost means ubuntu20 server is down

Rule ID: 202, 203

Level: 7, 9

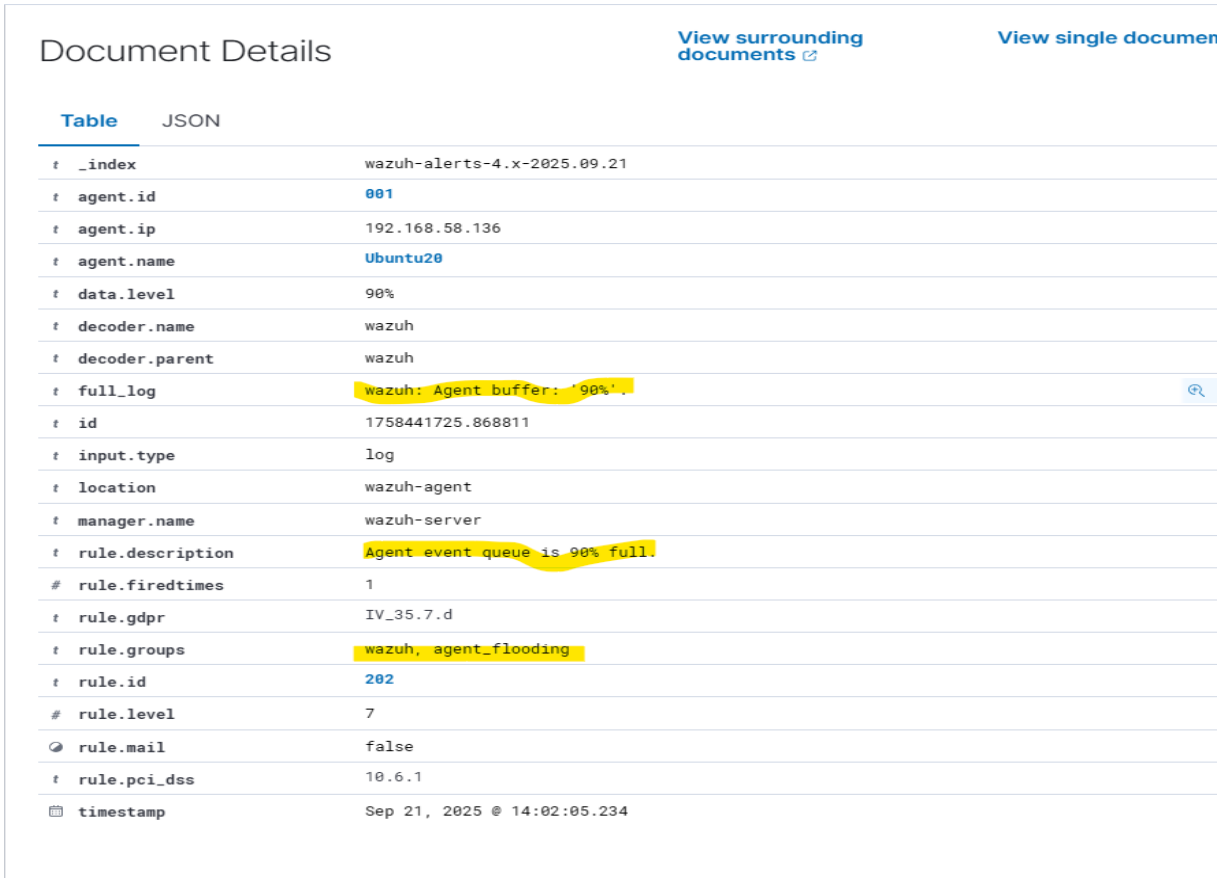


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Table JSON

_index	wazuh-alerts-4.x-2025.09.21
agent.id	001
agent.ip	192.168.58.136
agent.name	Ubuntu20
data.level	90%
decoder.name	wazuh
decoder.parent	wazuh
full_log	Wazuh: Agent buffer: '90%'.
id	1758441725.868811
input.type	log
location	wazuh-agent
manager.name	wazuh-server
rule.description	Agent event queue is 90% full.
rule.firedtimes	1
rule.gdpr	IV_35.7.d
rule.groups	wazuh, agent_flooding
rule.id	202
rule.level	7
rule.mail	false
rule.pci_dss	10.6.1
timestamp	Sep 21, 2025 @ 14:02:05.234

Here, Suricata detect:
Wazuh: Agent buffer: 90%
Agent event queue is 90% full
Rule.level: 07

Chapter 6

Result & Discussion

6.1 Attacking Machine

We assign attacking machine as Kali Linux, which IP address is 192.168.58.133. The attacker attack the system-

- Search the most vulnerable machine using Enumeration or port scanning using Nmap tool.
- Attacker try to enter the Ubuntu20
- Hacker perform several attack on that Ubuntu20. Such as, Brute Force Attack, DDOS attack

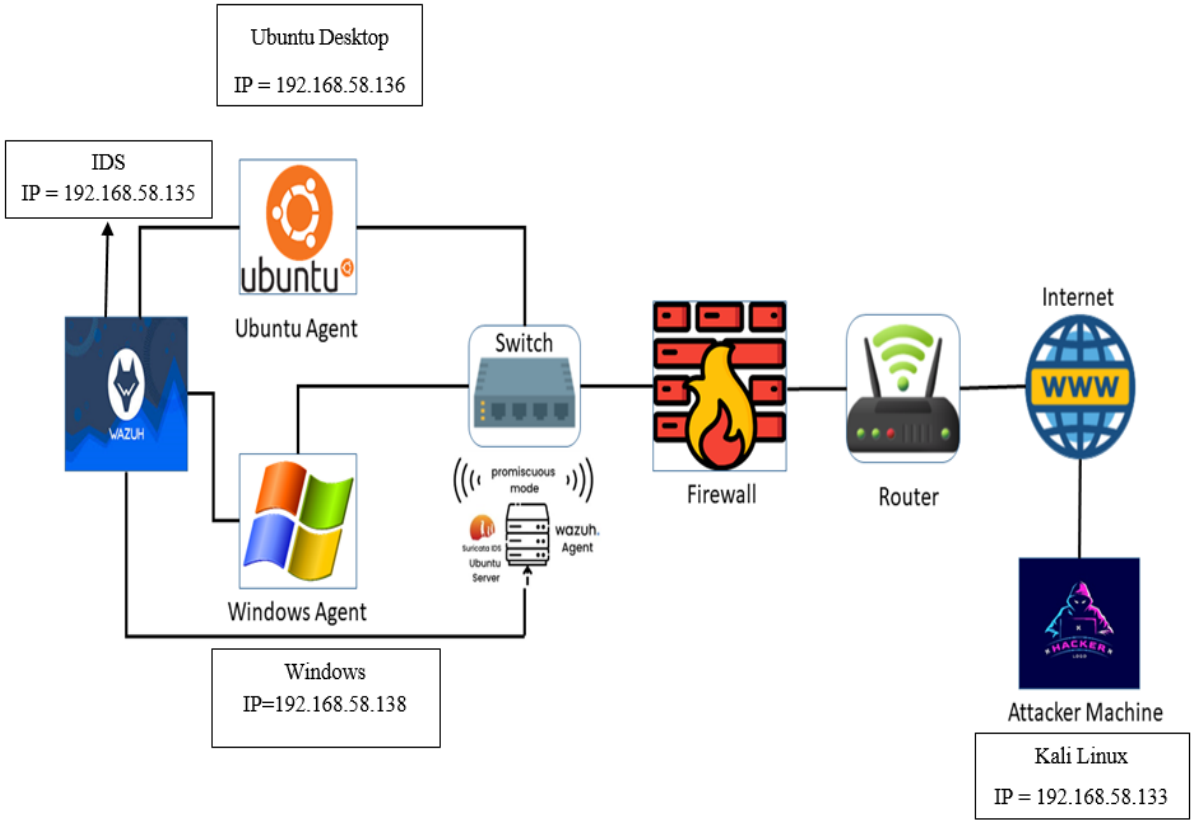


Figure 3.1 Implemented Design and Implementation Security Operations Center (SOC) Using Open Source Technologies SIEM for Industries

6.2 Network Intrusion Detection System (NIDS)

Suricata is open-source network intrusion detection and prevention system (IDS/IPS). It is intended to monitor network traffic and detect a variety of threats, including malware, intrusion attempts, and network anomalies. Using a rule-based language, Suricata analyzes network packets in real time, allowing it to identify and respond to suspicious or malicious activities. The non-profit organization OISF (Open Information Security Foundation) owns and develops Suricata

- Monitor network traffic used Promiscuous mode and detect a variety of threats, intrusion attempts and network anomalies
- Suricata analyzes network packets in real time
- Detect all of attacker activity.
- Show alert against those attack.
- Provide the all event and logs of hacker activity.
- Detect the abnormal behavior of all agents.

In conclusion, With this Design and Implementation Security Operations Center (SOC) Using Open Source Technologies for Induristies we can capture network packets from switch or NIC, capture the hacker activity for future research, and identify properly the hacker thinking.

Chapter 7

Conclusion

A “Design and Implementation Security Operations Center (SOC) Using Open Source Technologies for Induristies” outcome depends on a number of variables and the particular objectives for its deployment. Here are some important factors to think about before coming to a decision:

7.1 Enhanced Threat Detection

“Design and Implementation Security Operations Center (SOC) Using Open Source Technologies for Induristies” are intended to lure and trick potential attackers, enabling security experts to closely observe their actions and obtain insightful information about their strategies, methods, and motivations. This may result in earlier threat identification and a better comprehension of the changing threat landscape.

7.2 Early Warning System

The Early Warning System (EWS) in this project plays a vital role in detecting cyber threats at an early stage before they escalate into serious incidents. It is implemented by integrating Suricata Intrusion Detection System (IDS) with the Wazuh SIEM platform. Suricata operates in promiscuous mode on the monitoring interface and captures all inbound and outbound traffic. Using both predefined and custom rules, Suricata identifies suspicious activities such as port scanning with Nmap, brute-force login attempts with Hydra, and Distributed Denial of Service (DDoS) traffic. When such abnormal traffic is detected, Suricata generates alerts and forwards them to the Wazuh server. Wazuh then correlates these alerts with host-level data including authentication logs, system changes, and policy violations to produce enriched security events. These events are displayed in real-time on the Wazuh Dashboard for SOC analysts to review. By combining network-level and host-level monitoring, the EWS ensures better accuracy and reduces false positives. The system also supports threshold-based alerting, allowing repeated malicious attempts to be grouped as a single meaningful warning. With continuous monitoring and instant notifications, the EWS enables security teams to respond quickly to potential attacks. It provides visibility into early signs of compromise, strengthens defensive posture, and minimizes damage. Overall, the Early Warning System enhances the SOC by transforming raw data into actionable intelligence, thereby improving cyber resilience.

7.3 Limitations

One limitation of this project is that the Security Operations Center (SOC) relies mainly on open source tools such as Suricata and Wazuh. While powerful, these tools may lack some advanced enterprise-grade features, automation or vendor support compared to commercial SIEM solutions.

Another limitation is the resource dependency of IDS/IPS systems. Suricata, when monitoring high volumes of traffic in promiscuous mode, requires significant CPU and memory resources which may affect detection performance in large scale industrial environments.

The system also depends on the accuracy of rules and signatures. Custom rule creation can reduce false positives but may also miss new or sophisticated attacks if not regularly updated limiting the effectiveness of threat detection.

Finally, the lab-based implementation may not fully replicate the complexity of real-world industrial networks. Factors like encrypted traffic, insider threats, or large-scale distributed infrastructures are not fully addressed within this project setup

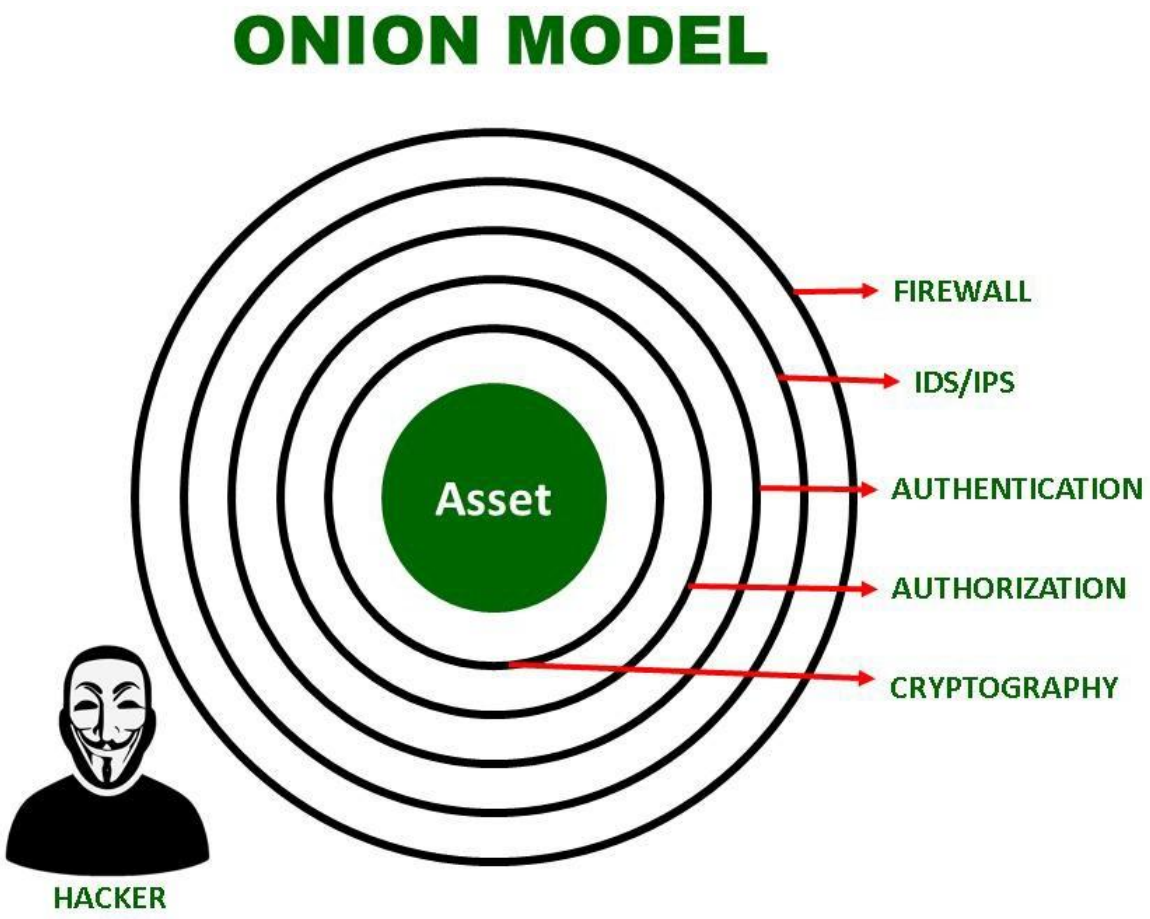
Chapter 8

Future Work

In future, We implement this “Design and Implementation Security Operations Center (SOC) Using Open Source Technologies for Induristies” with Security Onion instead of Wazuh.

8.1 Security Onion

Security Onion is a free and open platform for log management, enterprise security monitoring, and threat hunting is called Security Onion. It contains not only third-party tools like Playbook, FleetDM, osquery, CyberChef, Elasticsearch, Logstash, Kibana, Suricata, Zeek, and Wazuh but also its own tools for Alerts, Dashboards, Hunt, PCAP, and Cases. Security teams all across the world use Security Onion to monitor and safeguard their organizations, with over 2 million downloads to date. In security Onion Model there have 5 layer need to pass for the attacker for launch a successful attack. So, this will be more efficient compare with the Wazuh model.



8.2 Features of security onion

An effective open-source platform for intrusion detection and network security monitoring is called Security Onion. It combines several security techniques and technologies to offer a complete answer for identifying, evaluating, and addressing security risks. Security Onion has a number of important properties, such as:

- **Network Security Monitoring (NSM):** Security Onion offers a variety of NSM tools, including Suricata, Zeek (formerly Bro), and Snort. These technologies do real-time network traffic analysis in order to spot and warn about potential security breaches, intrusions, and suspicious activity.
- **Security Onion uses a variety of intrusion detection systems (IDS):** engines, such as Suricata and Snort, to identify known attack signatures and patterns in network traffic. It aids in the detection of malicious behavior, attempts at unauthorized access, and network-based assaults.
- **Packet Capture and Analysis:** For the purpose of capturing and analyzing network packets, Security Onion incorporates programs like Wireshark and NetworkMiner. It enables thorough examination of network traffic, obtaining useful data, and spotting possible security risks.
- **Advanced Analytics and Visualization:** Using applications like Kibana, Security Onion offers strong analytics and visualization capabilities. In order to help identify patterns, anomalies, and possible security problems, it enables the development of personalized dashboards, graphs, and reports.
- **Flexibility and Scalability:** Security Onion is made to be flexible and scalable, enabling implementation in settings of various sizes. It has the ability to interface with current security infrastructure and may be customized to meet unique corporate requirements.
- **Active Defense Capabilities:** Security Onion provides active defense capabilities with tools like CyberChef, a flexible data manipulation tool, as well as the ability to set up automatic response actions in response to identified security events.

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